P 6.1

$$\vec{P} = \vec{D} - \varepsilon_0 \vec{E} = (\varepsilon - \varepsilon_0)\vec{E}$$

$$\vec{M} = \frac{\vec{B}}{\mu_0} - \vec{H} = \left(\frac{1}{\mu_0} - \frac{1}{\mu}\right)\vec{B}$$

Idea: express induced current via $\sigma_{ij} E_j$

$$\vec{J} = -i\omega \vec{P} + i\vec{k}\times\vec{M} = -i\omega(\varepsilon-\varepsilon_0)\vec{E} + i\vec{k}\times\vec{B}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \vec{k}\times\vec{E} = \omega\vec{B}$$

$$\cancel{\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -\frac{\partial}{\partial t}\vec{\nabla}\times\vec{B}}$$

$$\cancel{i\vec{k}\times(i\vec{k}\times\vec{E}) = i\omega\, i\vec{k}\times\vec{B}}$$

$$\Rightarrow \vec{k}\times\vec{B} = \frac{1}{\omega}\vec{k}\times(\vec{k}\times\vec{E}) = \frac{1}{\omega}\left(\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}\right)$$

$$\vec{J} = -i\omega(\varepsilon-\varepsilon_0)\vec{E} + \frac{i}{\omega}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)\left(\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}\right)$$

$$J_i = -i\omega(\varepsilon-\varepsilon_0)E_i + \frac{i}{\omega}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)\left(k_i k_j E_j - k^2 E_i\right)$$

$$= \sigma_{ij} E_j$$

$$\sigma_{ij} = -i\omega(\varepsilon-\varepsilon_0)\delta_{ij} - i\frac{k^2}{\omega}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)\left(\delta_{ij} - \frac{k_i k_j}{k^2}\right)$$

$$K_{ij} = \delta_{ij} + \frac{i}{\omega\varepsilon_0}\sigma_{ij} \quad \text{(eq. 6.17)}$$

$$= \delta_{ij} + \frac{\varepsilon-\varepsilon_0}{\varepsilon_0}\delta_{ij} + \frac{k^2}{\varepsilon_0\omega^2}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)\left(\delta_{ij} - \frac{k_i k_j}{k^2}\right)$$

$$= \frac{\varepsilon(\omega)}{\varepsilon_0}\delta_{ij} + \frac{1}{\varepsilon_0\mu_0}\frac{k^2}{\omega^2}\left(1 - \frac{\mu_0}{\mu(\omega)}\right)\left(\delta_{ij} - \frac{k_i k_j}{k^2}\right)$$

But $c^2 = \frac{1}{\varepsilon_0\mu_0}$ $\Rightarrow$ we obtain (6.18) from the textbook on p. 74

P6.3 (a) $m_e \frac{d\vec{v}}{dt} = -e\vec{E} - \nu_e m_e \vec{v}$

let $\vec{v}(t) = \vec{v}(\omega) e^{-i\omega t}$, $\vec{E}(t) = \vec{E}(\omega) e^{-i\omega t}$

$$\Rightarrow \frac{d\vec{v}}{dt} = -i\omega \vec{v}(\omega) e^{-i\omega t}$$

$$\Rightarrow -i\omega m_e \vec{v}(\omega) e^{-i\omega t} = -e\vec{E}(\omega) e^{-i\omega t} - \nu_e m_e \vec{v}(\omega) e^{-i\omega t}$$

$$(i\omega - \nu_e) m_e \vec{v}(\omega) = e\vec{E}(\omega)$$

$$\vec{J}_{md}(\omega) = -e n_e \vec{v}(\omega) = -\frac{e^2 n_e \vec{E}(\omega)}{(i\omega - \nu_e) m_e} =$$

$$= \frac{i e^2 n_e}{(\omega + i\nu_e) m_e} \vec{E}(\omega) \Rightarrow \sigma_{ij}(\omega) = \delta_{ij} \frac{i e^2 n_e}{m_e(\omega + i\nu_e)}$$

$$K_{ij}(\omega) = \delta_{ij} + \frac{i}{\varepsilon_0 \omega} \sigma_{ij}(\omega) = \delta_{ij}\left(1 + \frac{i}{\varepsilon_0 \omega} \frac{i e^2 n_e}{m_e(\omega + i\nu_e)}\right)$$

$$= \delta_{ij}\left(1 - \frac{\omega_p^2}{\omega(\omega + i\nu_e)}\right) \quad \text{where } \omega_p^2 = \frac{e^2 n_e}{\varepsilon_0 m_e}$$

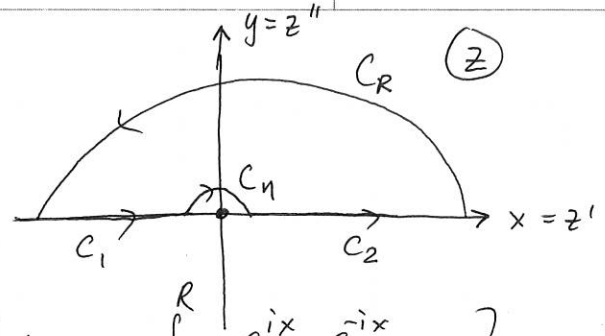
(b) $\sigma_{ij}(\omega) = \delta_{ij}\, \sigma(\omega)$ where $\sigma(\omega) = \frac{i e^2 n_e}{m_e(\omega + i\nu_e)}$

$$\sigma(\omega \to 0) = \frac{e^2 n_e}{m_e \nu_e} = \varepsilon_0 \frac{e^2 n_e}{\varepsilon_0 m_e} \frac{1}{\nu_e} = \frac{\varepsilon_0 \omega_p^2}{\nu_e}$$

$$\Rightarrow \eta = \frac{1}{\sigma(\omega = 0)} = \frac{\nu_e}{\varepsilon_0 \omega_p^2}.$$

P3

$$I_1 = \int_{-\infty}^{\infty} dx \, \frac{\sin x}{x}$$



$$I_1 = \lim_{\substack{\eta \to 0 \\ R \to \infty}} \left\{ \int_{-R}^{-\eta} \frac{e^{ix} - e^{-ix}}{2ix} dx + \int_{\eta}^{R} \frac{e^{ix} - e^{-ix}}{2ix} dx \right\}$$

Note that $\frac{\sin x}{x} \to 1$ for $x \to 0 \Rightarrow$ the integrand is not singular.

$$\int_{-R}^{-\eta} dx \, \frac{e^{-ix}}{x} = (x = -y) = -\int_{R}^{\eta} dy \, \frac{e^{iy}}{-y} = -\int_{\eta}^{R} dy \, \frac{e^{iy}}{y}$$

$$\int_{\eta}^{R} dx \, \frac{e^{-ix}}{x} = (x = -y) = -\int_{-\eta}^{-R} dy \, \frac{e^{iy}}{-y} = -\int_{-R}^{-\eta} dy \, \frac{e^{iy}}{y}$$

$$\Rightarrow I_1 = \lim_{\substack{\eta \to 0 \\ R \to \infty}} \left(\int_{-R}^{-\eta} \frac{e^{ix}}{ix} dx + \int_{\eta}^{R} \frac{e^{ix}}{ix} dx \right)$$

Consider now auxiliary integral

$$J_1 = \oint_{C = C_1 + C_\eta + C_2 + C_R} \frac{e^{iz}}{z} dz$$

where $C_1 = (-\infty, -\eta)$, $C_2 = (\eta, +\infty)$, C_R = circle of radius $R \to \infty$

C_η = circle of radius $\eta \to 0$

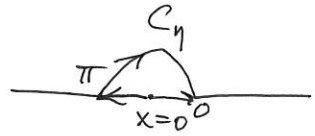
But $\frac{e^{iz}}{z}$ encloses no poles inside C, and is perfectly convergent on C_R

$\Rightarrow J_1 = 0$

This means $\int_{-R}^{-\eta} \frac{e^{ix}}{x} dx + \int_{\eta}^{R} \frac{e^{ix}}{x} dx = -\int_{C_\eta} \frac{e^{iz}}{z} dz - \int_{C_R} \frac{e^{iz}}{z} dz$

But $\int_{C_R} \frac{e^{iz}}{z}\,dz = 0$ since $e^{i(x+iy)} = e^{ix}e^{-y} \xrightarrow[y\to\infty]{} 0$

Therefore $iI_1 = -\int_{C_\eta} \frac{e^{iz}}{z}\,dz$



On C_η $z = \eta e^{i\varphi}$

$$I_1 = i\int_\pi^0 \frac{e^{i\eta e^{i\varphi}}}{\eta e^{i\varphi}}\, i\eta e^{i\varphi}\,d\varphi = -\int_\pi^0 d\varphi\,\underbrace{\left(1 + i\eta e^{i\varphi} + \dots\right)}_{\text{expand } e^{iz}} =$$

$= -(-\pi)$ + terms that vanish in the $\eta \to 0$ limit.

$$\Rightarrow I_1 = \int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx = \pi$$

Next $I_2 = \int_{-\infty}^{\infty} \frac{\sin^2(x)}{x^2}\,dx = \frac{1}{2}\,\mathrm{Re}\int_{-\infty}^{\infty} \frac{1-e^{i2x}}{x^2}\,dx$

$$\sin^2 x = \frac{1}{2}\left(1-\cos 2x\right) = \frac{1}{2}\,\mathrm{Re}\left(1 - e^{i2x}\right)$$

Hence consider $J_2 = \frac{1}{2}\,\mathrm{Re}\oint_C \frac{1-e^{i2z}}{z^2}\,dz = 0$

same contour as for J_1

$$\Rightarrow I_2 = \frac{1}{2}\,\mathrm{Re}\left(\int_{-R}^{-\eta} dx\,\frac{1-e^{i2x}}{x^2} + \int_\eta^R \frac{1-e^{i2x}}{x^2}\,dx\right)$$

$$= \frac{1}{2}\,\mathrm{Re}\left(-\int_{C_\eta} \frac{1-e^{i2z}}{z^2}\,dz - \underbrace{\int_{C_R} \frac{1-e^{i2z}}{z^2}\,dz}_{0 \text{ as before.}}\right)$$

$$\int_{C_\eta} \frac{1-e^{i2z}}{z^2}\,dz = \int_{C_\eta} \frac{1-\left(1+i2z-4z^2/4+\dots\right)}{z^2}\,dz =$$

$z = \eta e^{i\varphi}$

$$= -2i\int_{C_\eta} \frac{dz}{z} = -2i\int_\pi^0 \frac{i\eta e^{i\varphi}\,d\varphi}{\eta e^{i\varphi}} = 2\int_\pi^0 d\varphi = -2\pi \Rightarrow$$

$$I_2 = -\frac{1}{2}\,\mathrm{Re}\left(\int_{C_\eta} \frac{1-e^{i2z}}{z^2}\right) = -\frac{1}{2}(-2\pi) = \pi. \Rightarrow I_2 = \pi$$

as well!